

SUMMARY

Software Engineer with 6+ years of experience building backend systems, real-time applications, and data-driven solutions across automotive, research, and immersive environments. Strong expertise in C#, Python, and API-driven architectures, with hands-on experience in Android Automotive, simulation systems, and data pipelines. Proven ability to design scalable systems, integrate heterogeneous technologies, and deliver performant, maintainable software in cross-functional teams.

SKILLS & TECHNOLOGIES

- **Programming Languages:** C#, Python, Java, Kotlin, JavaScript, TypeScript, Ruby, C++
- **Frameworks & Libraries:** REST APIs, FastAPI, Flask, Ruby on Rails, gRPC, Unity, Vue.js
- **Data & Databases:** SQL, PostgreSQL, MySQL, MongoDB, Data Pipelines, ETL Concepts
- **Development Tools & IDE:** Google Colab, Unity3D, Visual Studio, VS Code, IntelliJ, Android Studio, Postman, Docker
- **Version Control & CI/CD:** GitHub, GitLab, Bitbucket, Jenkins
- **Systems & Domains:** Real-time systems, simulation, Android Automotive, backend integration
- **Concepts:** Clean Architecture, OOP, Modular Design Patterns, Agile/Scrum
- **Soft Skills:** Self Learning, Team Collaboration, Problem Solving, Analytical Thinking, Process Improvement
- **Languages:** English (B2-C1), German (A2-B1)

WORK EXPERIENCE

Master Thesis – Android Developer

Mar 2025 – Nov 2025

Mercedes-Benz AG

Stuttgart, Germany

- Developed an Android Automotive application using Kotlin, Coroutines, and Flow with a strong focus on performance, stability, and clean architecture.
- Implemented real-time data processing pipelines connected to vehicle signals and external data sources via structured APIs.
- Integrated low-level vehicle and hardware-related data (e.g. CAN bus signals) into Android applications.
- Collaborated closely with UX/UI designers to deliver intuitive, responsive user interfaces for large infotainment displays.
- Ensured code quality through modular design, debugging, profiling, and structured testing approaches.

Skills: Kotlin, Android Automotive OS, REST APIs, Hardware Integration, MVVM, Performance Optimization, Git, Docker, Linux

Research Assistant – C# Developer (Backend & Unity)

Oct 2023 – Mar 2026

Technische Universität Chemnitz

Chemnitz, Germany

- Developed C# based software components using object-oriented design and modular architectures.
- Implemented backend-facing features to collect, process, and persist simulation telemetry in SQL based systems.
- Designed REST style data exchange between Unity (C#) and Python services via FastAPI.
- Contributed to clean architecture discussions, refactoring, and internal code reviews.
- Worked closely with researchers to translate functional requirements into technical solutions.

Skills: C#, Unity3D, REST APIs, SQL, FastAPI, Backend Integration, Clean Code, Git

Mandatory Internship – Android Developer

Apr 2024 – Sep 2024

AUDI AG

Ingolstadt, Germany

- Built Android applications with Java/Kotlin using a FastAPI backend; implemented interactive UI elements, event driven logic, and efficient data communication aligning with scalable mobile app architecture.
- Implemented gRPC and REST APIs to streamline data exchange between the frontend and backend; validated endpoints using Postman on Linux. Integrated features such as noise cancellation, speech-to-text translation, and wake word detection.
- Designed and integrated AIDL interfaces for inter process communication within Android Automotive OS, enabling modular and efficient service interactions.
- Implemented wake word detection and speech-to-text translation features to support hands-free voice commands and improve user interaction.

Skills: Java, Kotlin, Python, FastAPI, Azure, gRPC, REST, Postman, BitBucket, Git, .Proto, Dagger dependency injection, Factory & MVVM Design pattern, Docker, Responsive UI

Student Assistant – Ruby on Rails Developer

Mar 2023 – Mar 2024

Fraunhofer IMW

Leipzig, Germany

- Developed backend modules and web crawlers using Ruby on Rails for a data-driven platform.
- Designed and integrated RESTful interfaces for internal and external data sources.
- Refactored existing codebases to improve maintainability and separation of concerns.
- Worked with SQL databases, optimizing queries and improving data processing performance.
- Used GitLab CI/CD pipelines for testing and deployment in Linux environments.

Skills: Ruby on Rails, REST APIs, SQL, Backend Development, GitLab CI/CD, Linux

Student Assistant – Full-Stack Developer

Fraunhofer IVV

Jun 2022 – Jan 2023
Dresden, Germany

- Developed backend services using Python (Flask) and REST APIs supporting frontend applications.
- Implemented database access logic and API endpoints consumed by Vue.js frontend components.
- Containerized backend services using Docker and supported CI workflows.
- Participated in code reviews and agile development processes.

Skills: Python, Flask, Vue.js, TypeScript, JavaScript, REST APIs, SQL, Docker, Git, Agile

Senior 3D & VR Game Developer - Unity (C#) Developer

Aug 2017 – Mar 2022

Abstron Pvt. Ltd

Lahore, Pakistan

- Developed cutting-edge VR, AR, and interactive simulations for Oculus, Pico, HTC Vive, Android, and Windows platforms.
- Set up complete VR scenes including lighting, materials, animations, gameplay logic, and environment design.
- Developed complex interaction systems, VR UI, physics-based mechanics, and real-time simulations.
- Integrated optimized game-ready 3D assets into Unity, ensuring high frame rates, quality shading, and immersive visuals.
- Engineered multiplayer systems using UNET and custom networking (UDP/TCP) with low-latency synchronization.
- Delivered customer-facing demos, gathered requirements, provided technical consulting, and led development cycles.
- Optimized rendering performance using Unity Profiler, batching, memory profiling, and GPU/CPU diagnostics.
- Managed cross-platform builds and deployments across Oculus, Android, iOS, and standalone environments.

Skills: Unity3D, C#, VR/AR/XR, Scene Setup, Lighting, Texturing, Shaders, Multiplayer, Performance Optimization, Dependency Injection

EDUCATION

Master's in Automotive Software Engineering

2022 – 2026

Technische Universität Chemnitz, Germany

Grade: 2,3

Bachelor of Science in Software Engineering

2011 – 2015

International Islamic University Islamabad, Pakistan

Grade: 1,7

PROJECTS

Fingerprints and Signature Verification System

Feb 2015 – Aug 2015

Final Year Project, Desktop Application (C#)

- Developed identity verification modules using advanced image processing and matching algorithms.
- Achieved a 97% accuracy rate, outperforming the university's existing system by 25%.

CERTIFICATES AND SEMINARS

Coding Hackathons

Transatlantic AI Hackathon: Participant

GABA Southern California, April, 2023

WorldInnovationDay Hack 2022: Participant

Hackmakers, May, 2022

Sustainable Supply Chain Deephack: Participant

Ultrahack, April, 2022

FormulaAI Hack 2022: Participant

Hackmakers, March, 2022

Certificates

Certificate in Programming Essentials in Python

Cisco Networking Academy, March, 2022

Introduction to Data Visualization with Matplotlib

DataCamp, April, 2022

Cleaning Data in Python

DataCamp, May, 2022

Reshaping Data with Pandas

DataCamp, June, 2022

RESEARCH AND VOLUNTEERING

Data Analyst — Machine Learning Learner

Aug 2020 – Mar 2022

ICodeGuru, USA

- ICodeGuru is an initiative led by Silicon Valley industry experts to mentor students and prepare them for top tech companies.
- Worked on Data Analysis techniques and methods using Python.
- Programming and experimentation on Jupyter Notebook.
- Built and evaluated different machine learning models using libraries and frameworks such as scikit-learn, PyTorch, Keras, OpenCV, TensorFlow.
- Completed 8 DataCamp courses on Data Science.
- Participated in 4 international coding hackathons on Data Analysis and Machine Learning.
- Performed data transformation from local databases to Salesforce and AWS using Mulesoft API and deployed on Anypoint platform and also created basic dashboards and worksheets in Tableau Public and PowerBI.
- Delivered presentations on Data Analysis, Salesforce Single Sign-On, Authentication, and Authorization.

Skills: Python, Machine Learning, Mulesoft, Docker, Postman, Data Analysis, Jupyter Notebook, scikit-learn, PyTorch, Keras, TensorFlow, OpenCV, Salesforce, AWS, Mulesoft, Dashboards, Data Visualization, Hackathons